Exam 2: Part 1 (True/False and Multiple Choice). This part of exam 1 is not timed and is also open notes, open book, open Canva materials, with no outside assistance permitted (from others or from outside materials).

Question 1 — 2 pts

The pointer register X is referenced as the following 8-bit GPR pair.

- ○ R29 (XH) R28 (XL)
- ● R27 (XH) R26 (XL)
- ○ R30 (XH) R31 (XL)
- ○ R31 (XH) R30 (XL)
- ○ R26 (XH) R27 (XL)

Question 2 — 5 pts

Specify the addressing mode used (**given in bold**) in the instruction:

SUB **R23**, **R25**

- ○ Single Register/Immediate Value
- ● Register
- ○ Indirect Address
- ○ Direct Address

Question 3 — 2 pts

Specify the addressing mode used (**given in bold**) in the instruction:

BRNE **AGAIN**

- ○ Single Register/Immediate Value
- ○ Indirect Address
- ● Direct Address
- ○ Register

Question 4

2 pts

Specify the addressing mode used (**given in bold**) in the instruction:

IN R14, **PINC**

- ○ Single Register/Immediate Value
- ◕ Direct Address
- ○ Indirect Address
- ○ Register

Question 5

2 pts

Specify the addressing mode used (**given in bold**) in the instruction:

ST **Y+**, R27

- ○ Direct Address
- ○ Single Register/Immediate Value
- ◕ Indirect Address
- ○ Register

Question 6 — 2 pts

Specify the addressing mode used (**given in bold**) in the instruction:

LDI R31, **0x9B**

- ○ Register
- ○ Indirect Address
- ● Single Register/Immediate Value
- ○ Direct Address

Question 7 — 3 pts

Match the I/O register to its function.

I/O register used to configure port pins as inputs or outputs	[Choose]	DDRn
I/O register used to output values to the port pins	[Choose]	PORTn
I/O register used to store values read from the port pins	[Choose]	PINn
I/O register used to enable pull-up resistors for port pins configured as inputs	[Choose]	PORTn

For question 7: choices are PINn, PORTn, DDRn, SREG, SP

Question 8

2 pts

Specify the memory space(s) used by the operands in the instruction:

STS 0x87, R26

- ○ Code Memory
- ● Data Memory
- ○ Both

Question 9

2 pts

Specify the memory space(s) used by the operands in the instruction:

CALL DELAY

- ○ Data Memory
- ○ None
- ○ Both
- ● Code Memory

Question 10

2 pts

Specify the memory space(s) used by the operands in the instruction:

LPM R18, Z+

- ● Both
- ○ Code Memory
- ○ None
- ○ Data Memory

Question 11

2 pts

Specify the memory space(s) used by the operands in the instruction:

SUB R17, R16

- ○ None
- ● Data Memory
- ○ Both
- ○ Code Memory

Question 12

2 pts

Specify the memory space(s) used by the operands in the instruction:

BRNE LOOP

- ○ Data Memory
- ○ Both
- ● Code Memory
- ○ None

Question 13

2 pts

Specify the memory space(s) used by the operands in the instruction:

NOP

- ● None
- ○ Data Memory
- ○ Code Memory
- ○ Both

Question 14

3 pts

What is the duty cycle of the waveform given by the following instruction sequence?

```
SBI DDRD, 7
HERE:   CBI, PORTD, 7
        CALL DELAY
        SBI PORTD, 7
        CALL DELAY
        CALL DELAY
        CALL DELAY
    RJMP HERE
```

- ○ 25%
- ○ 66%
- ○ 20%
- ○ 80%
- ○ 33%
- ● 75%

Question 15

2 pts

The instructions SBIS and SBIC are both conditional branch instructions that can branch to a code memory location within 64 bytes of the current Program Counter (PC).

- ○ True
- ✗ False

SBIS and SBIC Skip if condition is met not branch to another code memory location

Question 16

3 pts

Determine the values of R16 and R17 (in binary) from the instruction sequence.

LDI R16, 0x58

LDI R17, 0x84

OR R16, R17

- ○ 1000 0100
- ○ 0101 1000
- ○ 0000 0000
- ○ 1101 0100
- ○ 1000 1100
- ✗ 1101 1100

Question 17

4 pts

Specify the register(s) that the product is stored and the values of those registers after executing the following instructions.

LDI R20, 0x6C

LDI R21, 0x7B

MUL R21, R20

- ○ R21 = 0x33 R20 = 0xE4
- ○ R21 = 0xE4 R20 = 0x33
- ○ R1 = 0xE4 R0 = 0x33
- ○ R21 = 0x33E4
- ○ R21 = 0xE4
- ✗ R1 = 0x33 R0 = 0xE4

Question 18

5 pts

Find the contents of R20 **in hex** and the value of the carry bit (C) after executing the following code sequence.

```
CLC
LDI R20, 0x9D
ROR R20
ROR R20
LSL R20
```

1001 1101 C = 0
0100 1110 C = 1
1010 0111 C = 0
0100 1110 C = 1

- ○ R20 = 0x4C C = 1
- ○ R20 = 0xE4 C = 1
- ○ R20 = 0x7A C = 1
- ● R20 = 0x4E C = 1
- ○ R20 = 0xA7 C = 1
- ○ R20 = 0xA7 C = 0

Question 19

3 pts

Determine the 8-bit code memory location associated with the ASCII character 's'.

```
; look up table for assembly language program
.ORG 0x0150
NewString: .DB  “this exam”
```

0x02A0 t
0x02A1 h
0x02A2 i
0x02A3 s

- ● 0x02A3
- ○ 0x0151
- ○ 0x0153
- ○ 0x0302
- ○ 0x0303
- ○ 0x02A1